

CHEATSHEET · QUICK REFERENCE

Lab 12 — Performance Monitoring, Baselining, Event Logs

Capture CPU, memory, disk-IOPS and network baselines with sysstat/fio, inspect and rotate event logs, and wire thresholds and alerts with node_exporter.

The four metric classes

```
uptime
```

Load averages (1/5/15 min)

```
mpstat 1 5
```

CPU: idle/user/system/iowait %

```
free -h ; vmstat 1 5
```

Memory: available, swap, paging

```
iostat -xz 1 5
```

Disk: %util, r/s+w/s (IOPS), await (latency)

```
sar -n DEV 1 5
```

Network throughput per interface

Generate load and baseline

```
fio --name=t --filename=/tmp/iotest --size=128M --bs=4k --rw=randwrite --ioengine=libaio --iodepth=
```

Disk load test

```
mkdir -p /var/log/baseline
```

Baseline directory

```
mpstat 1 3 > /var/log/baseline/baseline-$(date +%F).txt
```

Persist a baseline file

Event logs, rotation and shipping

```
journalctl --since "1 hour ago" -p err
```

Recent errors

```
journalctl -u nginx -n 50
```

Per-unit logs

```
echo '*.* @siem.lab.local:514' > /etc/rsyslog.d/90-forward.conf
```

Forward via TCP (@@)

Thresholds and alerting

```
systemctl enable --now prometheus-node-exporter
```

Expose metrics on :9100

```
curl -s http://127.0.0.1:9100/metrics | grep node_cpu_seconds_total
```

Scrape a metric

```
rate(node_cpu_seconds_total{mode!="idle"}[5m]) > 0.85
```

PromQL alert condition